PIMPRI CHINCHWAD EDUCATION TRUST's.
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
(An Autonomous Institute)
Affiliated to Savitribai Phule Pune University (SPPU)
ISO 21001:2018 Certified by TUV SUD

T.Y. B. TECH	**Year:** 2025 – 26	**Subject:** FSDL
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Branch: Computer Engineering		**Division: B**

Assignment:4

1. Objective

The primary objective of this assignment is to design and develop an interactive data visualization dashboard using HTML, CSS, and JavaScript. The goal is to present real-life data (such as weather information) in a clear and visually appealing manner using charts and graphs. Libraries like Chart.js or D3.js are used to create dynamic and interactive visualizations, while Bootstrap CDN helps in building a responsive layout.

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2. Learning Outcomes

After completing this assignment, the following skills and knowledge are gained:

- Understanding how to build interactive dashboards using JavaScript.
- Learning to visualize real-world data using charting libraries like Chart.js or D3.js.
- Applying HTML and CSS to design structured and visually appealing layouts.
- Using Bootstrap CDN to create responsive and user-friendly interfaces.
- Gaining experience in handling and representing data effectively.
- Improving analytical thinking by interpreting graphical data.
- Enhancing front-end development and UI/UX design skills.

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3. Brief Theory

A data visualization dashboard is an interface that displays data in graphical formats such as charts, graphs, and maps, making it easier to understand patterns, trends, and insights. Dashboards are widely used in real-life applications like weather monitoring, business analytics, and health tracking.

HTML is used to structure the dashboard layout, including containers for charts and information panels. CSS is used to style the dashboard, improving readability and visual appeal.

JavaScript plays a key role in making the dashboard interactive by fetching, processing, and dynamically displaying data. Libraries like Chart.js and D3.js simplify the creation of charts such as bar graphs, line charts, and pie charts. These libraries provide built-in functions to animate and update data in real-time.

Bootstrap is used to create a responsive layout with grid systems and pre-designed components, ensuring that the dashboard works smoothly across different devices and screen sizes.

Code: (html)

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>StormPulse — Live Weather Dashboard</title>
<script
src="https://cdn.jsdelivr.net/npm/chart.js@4.4.1/dist/chart.umd.min.js"></script>
<style>
  @import
url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap');

  * { margin: 0; padding: 0; box-sizing: border-box; }

  :root {
    --bg: #0b0f1a;
    --card: #131a2e;
    --card-hover: #1a2340;
    --accent: #38bdf8;
    --accent2: #a78bfa;
    --accent3: #f472b6;
    --accent4: #34d399;
    --text: #e2e8f0;
    --text-dim: #64748b;
    --border: #1e293b;
    --glow: rgba(56, 189, 248, 0.15);
  }

  body {
    font-family: 'Inter', sans-serif;
    background: var(--bg);
```

```
  color: var(--text);
  min-height: 100vh;
  overflow-x: hidden;
}

.particles {
  position: fixed; top: 0; left: 0; width: 100%; height: 100%;
  pointer-events: none; z-index: 0; overflow: hidden;
}
.particle {
  position: absolute;
  background: var(--accent);
  border-radius: 50%;
  opacity: 0;
  animation: float linear infinite;
}
@keyframes float {
  0% { opacity: 0; transform: translateY(100vh) scale(0); }
  10% { opacity: 0.6; }
  90% { opacity: 0.6; }
  100% { opacity: 0; transform: translateY(-10vh) scale(1); }
}

header {
  position: relative; z-index: 1;
  padding: 24px 40px;
  display: flex; align-items: center; justify-content: space-between;
  border-bottom: 1px solid var(--border);
  backdrop-filter: blur(12px);
  background: rgba(11, 15, 26, 0.8);
}
.logo {
  display: flex; align-items: center; gap: 12px;
  font-size: 22px; font-weight: 700; letter-spacing: -0.5px;
}
.logo svg { filter: drop-shadow(0 0 8px var(--accent)); }
.logo span { color: var(--accent); }

.city-selector {
  display: flex; align-items: center; gap: 12px;
}
.city-selector select {
  background: var(--card); color: var(--text); border: 1px solid var(--border);
  padding: 8px 16px; border-radius: 8px; font-size: 14px; font-family: inherit;
  cursor: pointer; outline: none; transition: border-color 0.2s;
}
.city-selector select:hover { border-color: var(--accent); }
.live-dot {
```

```
  width: 8px; height: 8px; background: #22c55e; border-radius: 50%;
  animation: pulse-dot 2s ease infinite;
}
@keyframes pulse-dot {
  0%, 100% { box-shadow: 0 0 0 0 rgba(34,197,94,0.5); }
  50% { box-shadow: 0 0 0 6px rgba(34,197,94,0); }
}
.live-label { font-size: 12px; color: #22c55e; font-weight: 500; }

.dashboard {
  position: relative; z-index: 1;
  max-width: 1400px; margin: 0 auto;
  padding: 32px 40px;
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  grid-template-rows: auto auto auto;
  gap: 20px;
}

.card {
  background: var(--card);
  border: 1px solid var(--border);
  border-radius: 16px;
  padding: 24px;
  transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
  cursor: pointer;
  position: relative;
  overflow: hidden;
}
.card::before {
  content: '';
  position: absolute; top: 0; left: 0; right: 0; height: 2px;
  background: linear-gradient(90deg, transparent, var(--accent), transparent);
  opacity: 0; transition: opacity 0.3s;
}
.card:hover {
  background: var(--card-hover);
  transform: translateY(-4px);
  box-shadow: 0 12px 40px rgba(0,0,0,0.3), 0 0 30px var(--glow);
}
.card:hover::before { opacity: 1; }

.card-title {
  font-size: 12px; font-weight: 600; text-transform: uppercase;
  letter-spacing: 1.2px; color: var(--text-dim); margin-bottom: 12px;
}

.hero-card {
```

```
  grid-column: span 2;
  display: flex; flex-direction: column; justify-content: space-between;
  background: linear-gradient(135deg, #131a2e 0%, #1a1040 100%);
}
.hero-temp {
  font-size: 96px; font-weight: 200; line-height: 1;
  background: linear-gradient(135deg, var(--accent), var(--accent2));
  -webkit-background-clip: text; -webkit-text-fill-color: transparent;
}
.hero-condition { font-size: 20px; color: var(--text-dim); margin-top: 4px; }
.hero-details {
  display: grid; grid-template-columns: repeat(3, 1fr); gap: 16px; margin-top: 24px;
}
.hero-detail-item { text-align: center; }
.hero-detail-value { font-size: 24px; font-weight: 600; }
.hero-detail-label { font-size: 11px; color: var(--text-dim); margin-top: 2px; }
.hero-weather-icon { font-size: 64px; position: absolute; top: 24px; right: 32px; }

.stat-card .stat-value {
  font-size: 36px; font-weight: 600; margin-bottom: 4px;
}
.stat-card .stat-sub { font-size: 13px; color: var(--text-dim); }
.stat-card .stat-icon {
  position: absolute; top: 20px; right: 20px;
  width: 40px; height: 40px; border-radius: 10px;
  display: flex; align-items: center; justify-content: center;
  font-size: 18px;
}

.chart-card { grid-column: span 2; height: 320px; }
.chart-card canvas { height: 260px !important; }

.radar-card { grid-column: span 1; height: 320px; }
.radar-card canvas { height: 260px !important; }
.doughnut-card { grid-column: span 1; height: 320px; }
.doughnut-card canvas { height: 220px !important; }

.forecast-card {
  grid-column: span 4;
  display: flex; flex-direction: column;
}
.forecast-row {
  display: grid; grid-template-columns: repeat(7, 1fr); gap: 12px; margin-top: 16px;
}
.forecast-day {
  text-align: center; padding: 16px 8px;
  background: rgba(255,255,255,0.02);
  border-radius: 12px; border: 1px solid transparent;
```

```
  transition: all 0.25s;
}
.forecast-day:hover {
  border-color: var(--accent);
  background: rgba(56,189,248,0.06);
  transform: scale(1.05);
}
.forecast-day .day-name { font-size: 12px; color: var(--text-dim); font-weight: 600; }
.forecast-day .day-icon { font-size: 28px; margin: 8px 0; }
.forecast-day .day-high { font-size: 18px; font-weight: 600; }
.forecast-day .day-low { font-size: 13px; color: var(--text-dim); }

/* Popup overlay */
.popup-overlay {
  position: fixed; inset: 0;
  background: rgba(0,0,0,0.7);
  backdrop-filter: blur(8px);
  z-index: 1000;
  display: flex; align-items: center; justify-content: center;
  opacity: 0; pointer-events: none;
  transition: opacity 0.3s;
}
.popup-overlay.active { opacity: 1; pointer-events: all; }

.popup {
  background: var(--card);
  border: 1px solid var(--border);
  border-radius: 20px;
  padding: 0;
  max-width: 520px; width: 90%;
  transform: scale(0.9) translateY(20px);
  transition: transform 0.35s cubic-bezier(0.4, 0, 0.2, 1);
  overflow: hidden;
}
.popup-overlay.active .popup {
  transform: scale(1) translateY(0);
}
.popup-header {
  padding: 24px 28px 16px;
  display: flex; align-items: center; justify-content: space-between;
}
.popup-header h2 { font-size: 18px; font-weight: 600; }
.popup-close {
  width: 32px; height: 32px; border-radius: 8px;
  border: 1px solid var(--border); background: transparent;
  color: var(--text-dim); cursor: pointer; font-size: 16px;
  display: flex; align-items: center; justify-content: center;
  transition: all 0.2s;
```

```
}
.popup-close:hover { background: rgba(255,255,255,0.05); color: var(--text); }
.popup-body { padding: 0 28px 28px; }
.popup-body canvas { max-height: 240px; }
.popup-stats {
  display: grid; grid-template-columns: 1fr 1fr; gap: 12px; margin-bottom: 20px;
}
.popup-stat {
  background: rgba(255,255,255,0.03); padding: 14px; border-radius: 10px;
}
.popup-stat-label { font-size: 11px; color: var(--text-dim); text-transform:
uppercase; letter-spacing: 0.8px; }
.popup-stat-value { font-size: 22px; font-weight: 600; margin-top: 4px; }

/* Toast notification */
.toast {
  position: fixed; bottom: 32px; right: 32px;
  background: linear-gradient(135deg, #1e293b, #0f172a);
  border: 1px solid var(--accent);
  border-radius: 12px; padding: 16px 24px;
  z-index: 2000;
  display: flex; align-items: center; gap: 12px;
  transform: translateY(120px); opacity: 0;
  transition: all 0.4s cubic-bezier(0.4, 0, 0.2, 1);
  box-shadow: 0 8px 32px rgba(56,189,248,0.2);
}
.toast.show { transform: translateY(0); opacity: 1; }
.toast-icon { font-size: 20px; }
.toast-text { font-size: 13px; }
.toast-text strong { display: block; font-size: 14px; margin-bottom: 2px; }

/* Tooltip */
.tip {
  position: absolute; background: #1e293b; color: var(--text);
  font-size: 12px; padding: 8px 14px; border-radius: 8px;
  border: 1px solid var(--border);
  pointer-events: none; opacity: 0; transition: opacity 0.2s;
  white-space: nowrap; z-index: 500;
  box-shadow: 0 4px 12px rgba(0,0,0,0.4);
}
.tip::after {
  content: ''; position: absolute; bottom: -5px; left: 50%; transform: translateX(-
50%);
  border-left: 5px solid transparent; border-right: 5px solid transparent;
  border-top: 5px solid #1e293b;
}

@media (max-width: 900px) {
```

```
    .dashboard { grid-template-columns: 1fr 1fr; padding: 16px; }
    .hero-card, .chart-card, .forecast-card { grid-column: span 2; }
    .forecast-row { grid-template-columns: repeat(4, 1fr); }
    header { padding: 16px 20px; }
  }
  @media (max-width: 520px) {
    .dashboard { grid-template-columns: 1fr; }
    .hero-card, .chart-card, .forecast-card, .radar-card, .doughnut-card { grid-column:
span 1; }
    .forecast-row { grid-template-columns: repeat(3, 1fr); }
    .hero-temp { font-size: 64px; }
  }
</style>
</head>
<body>

<div class="particles" id="particles"></div>

<header>
  <div class="logo">
    <svg width="28" height="28" viewBox="0 0 24 24" fill="none"
stroke="#38bdf8" stroke-width="2" stroke-linecap="round">
      <path d="M13 2L3 14h9l-1 8 10-12h-9l1-8z"/>
    </svg>
    <span>Storm</span>Pulse
  </div>
  <div class="city-selector">
    <div class="live-dot"></div>
    <span class="live-label">LIVE</span>
    <select id="citySelect">
      <option value="mumbai">Mumbai, India</option>
      <option value="tokyo">Tokyo, Japan</option>
      <option value="london">London, UK</option>
      <option value="newyork">New York, USA</option>
      <option value="sydney">Sydney, Australia</option>
    </select>
  </div>
</header>

<div class="dashboard">
  <div class="card hero-card" onclick="openPopup('temperature')">
    <div>
      <div class="card-title">Current Weather</div>
      <div class="hero-temp" id="heroTemp">32°</div>
      <div class="hero-condition" id="heroCondition">Partly Cloudy</div>
    </div>
    <div class="hero-weather-icon" id="heroIcon">⛅</div>
    <div class="hero-details">
```

```
      <div class="hero-detail-item">
        <div class="hero-detail-value" id="feelsLike">35°</div>
        <div class="hero-detail-label">Feels Like</div>
      </div>
      <div class="hero-detail-item">
        <div class="hero-detail-value" id="heroHumidity">72%</div>
        <div class="hero-detail-label">Humidity</div>
      </div>
      <div class="hero-detail-item">
        <div class="hero-detail-value" id="heroPressure">1012</div>
        <div class="hero-detail-label">hPa</div>
      </div>
    </div>
  </div>

  <div class="card stat-card" onclick="openPopup('wind')">
    <div class="card-title">Wind Speed</div>
    <div class="stat-value" id="windSpeed" style="color:var(--accent)">18
km/h</div>
    <div class="stat-sub" id="windDir">NW — Gentle Breeze</div>
    <div class="stat-icon" style="background:rgba(56,189,248,0.12)">💨</div>
  </div>

  <div class="card stat-card" onclick="openPopup('uv')">
    <div class="card-title">UV Index</div>
    <div class="stat-value" id="uvValue" style="color:var(--accent3)">6</div>
    <div class="stat-sub" id="uvLabel">High — Wear Sunscreen</div>
    <div class="stat-icon" style="background:rgba(244,114,182,0.12)">☀️</div>
  </div>

  <div class="card stat-card" onclick="openPopup('visibility')">
    <div class="card-title">Visibility</div>
    <div class="stat-value" id="visValue" style="color:var(--accent4)">8.5 km</div>
    <div class="stat-sub">Good</div>
    <div class="stat-icon" style="background:rgba(52,211,153,0.12)">👁️</div>
  </div>

  <div class="card stat-card" onclick="openPopup('rain')">
    <div class="card-title">Rain Chance</div>
    <div class="stat-value" id="rainChance" style="color:var(--
accent2)">45%</div>
    <div class="stat-sub" id="rainSub">Light showers expected</div>
    <div class="stat-icon" style="background:rgba(167,139,250,0.12)">🌧️</div>
  </div>

  <div class="card chart-card">
    <div class="card-title">24-Hour Temperature Trend</div>
```

```
    <canvas id="tempChart"></canvas>
  </div>

  <div class="card chart-card">
    <div class="card-title">Hourly Precipitation & Humidity</div>
    <canvas id="precipChart"></canvas>
  </div>

  <div class="card radar-card">
    <div class="card-title">Air Quality Breakdown</div>
    <canvas id="aqiRadar"></canvas>
  </div>

  <div class="card doughnut-card" onclick="openPopup('aqi')">
    <div class="card-title">Overall AQI</div>
    <canvas id="aqiDoughnut"></canvas>
    <div style="text-align:center;margin-top:8px;font-size:13px;color:var(--text-
dim)" id="aqiLabel">Moderate — 128</div>
  </div>

  <div class="card forecast-card">
    <div class="card-title">7-Day Forecast</div>
    <div class="forecast-row" id="forecastRow"></div>
  </div>
</div>

<div class="popup-overlay" id="popupOverlay" onclick="closePopup(event)">
  <div class="popup" onclick="event.stopPropagation()">
    <div class="popup-header">
      <h2 id="popupTitle">Details</h2>
      <button class="popup-close" onclick="closePopup()">&times;</button>
    </div>
    <div class="popup-body" id="popupBody"></div>
  </div>
</div>

<div class="toast" id="toast">
  <span class="toast-icon"> ⚡ </span>
  <span class="toast-text" id="toastText"><strong>Weather Alert</strong>UV
index rising — stay hydrated!</span>
</div>

<div class="tip" id="tooltip"></div>

<script>
const CITIES = {
  mumbai: {
```

```
    name: 'Mumbai', temp: 32, feelsLike: 35, condition: 'Partly Cloudy', icon: '⛅',
    humidity: 72, pressure: 1012, wind: 18, windDir: 'NW', uv: 6, vis: 8.5, rain: 45,
    aqi: 128, aqiLabel: 'Moderate',
    hourlyTemp:
[28,27,27,26,26,27,28,30,31,32,33,34,34,33,33,32,31,30,29,29,28,28,27,27],
    hourlyPrecip: [0,0,5,10,15,5,0,0,0,0,10,20,35,40,30,15,5,0,0,0,5,10,5,0],
    hourlyHumidity:
[78,80,82,84,83,80,76,72,68,65,63,64,68,72,74,73,70,68,70,73,76,78,79,80],
    aqiBreakdown: [42,68,85,55,38,30],
    forecast: [
      {day:'Mon',icon:'⛅',high:33,low:26},{day:'Tue',icon:'🌧',high:30,low:25},
      {day:'Wed',icon:'⛈',high:28,low:24},{day:'Thu',icon:'🌧',high:29,low:25},
      {day:'Fri',icon:'☀',high:34,low:27},{day:'Sat',icon:'☀',high:35,low:28},
      {day:'Sun',icon:'⛅',high:33,low:27}
    ]
  },
  tokyo: {
    name: 'Tokyo', temp: 18, feelsLike: 16, condition: 'Clear Sky', icon: '☀',
    humidity: 55, pressure: 1018, wind: 12, windDir: 'E', uv: 4, vis: 12, rain: 10,
    aqi: 62, aqiLabel: 'Good',
    hourlyTemp:
[14,13,13,12,12,13,15,16,17,18,19,20,20,19,19,18,17,16,15,15,14,14,13,13],
    hourlyPrecip: [0,0,0,0,0,0,0,0,0,0,5,5,0,0,0,0,0,0,0,0,0,0,0,0],
    hourlyHumidity:
[62,65,67,70,68,64,58,55,52,50,48,49,52,56,58,60,62,65,67,68,64,62,61,63],
    aqiBreakdown: [20,30,25,18,12,10],
    forecast: [
      {day:'Mon',icon:'☀',high:19,low:12},{day:'Tue',icon:'⛅',high:17,low:11},
      {day:'Wed',icon:'🌧',high:14,low:10},{day:'Thu',icon:'☀',high:20,low:13},
      {day:'Fri',icon:'☀',high:22,low:14},{day:'Sat',icon:'⛅',high:18,low:12},
      {day:'Sun',icon:'☀',high:21,low:14}
    ]
  },
  london: {
    name: 'London', temp: 11, feelsLike: 8, condition: 'Overcast', icon: '☁',
    humidity: 82, pressure: 1005, wind: 24, windDir: 'SW', uv: 2, vis: 6, rain: 70,
    aqi: 75, aqiLabel: 'Moderate',
    hourlyTemp: [9,8,8,7,7,8,9,10,11,11,12,12,12,11,11,10,10,9,9,8,8,8,7,7],
    hourlyPrecip:
[20,25,30,35,30,20,15,10,5,10,20,35,45,50,40,30,20,15,20,25,30,25,20,15],
    hourlyHumidity:
[85,87,88,90,88,85,82,80,78,80,82,85,88,90,88,85,83,82,84,86,88,87,86,85],
    aqiBreakdown: [28,40,35,25,20,15],
    forecast: [
      {day:'Mon',icon:'🌧',high:11,low:6},{day:'Tue',icon:'🌧',high:10,low:5},
      {day:'Wed',icon:'☁',high:9,low:4},{day:'Thu',icon:'⛅',high:12,low:7},
```

```
    {day:'Fri',icon:'☀️',high:14,low:8},{day:'Sat',icon:'☁️',high:13,low:7},
    {day:'Sun',icon:'🌧️',high:10,low:5}
  ]
 },
 newyork: {
  name: 'New York', temp: 22, feelsLike: 23, condition: 'Sunny', icon: '☀️',
  humidity: 48, pressure: 1020, wind: 15, windDir: 'SE', uv: 7, vis: 14, rain: 5,
  aqi: 52, aqiLabel: 'Good',
  hourlyTemp:
[18,17,16,16,17,18,19,20,21,22,23,24,24,24,23,22,21,20,19,19,18,18,17,17],
  hourlyPrecip: [0,0,0,0,0,0,0,0,0,0,0,0,0,5,5,0,0,0,0,0,0,0,0,0],
  hourlyHumidity:
[55,58,60,62,60,56,52,48,45,43,42,40,41,44,47,50,52,55,58,60,58,56,55,56],
  aqiBreakdown: [18,22,20,15,10,8],
  forecast: [
    {day:'Mon',icon:'☀️',high:24,low:17},{day:'Tue',icon:'☀️',high:25,low:18},
    {day:'Wed',icon:'☁️',high:22,low:16},{day:'Thu',icon:'🌧️',high:18,low:14},
    {day:'Fri',icon:'☁️',high:20,low:15},{day:'Sat',icon:'☀️',high:23,low:17},
    {day:'Sun',icon:'☀️',high:26,low:19}
  ]
 },
 sydney: {
  name: 'Sydney', temp: 24, feelsLike: 25, condition: 'Breezy & Warm', icon: '🌤️',
  humidity: 60, pressure: 1015, wind: 22, windDir: 'NE', uv: 8, vis: 10, rain: 20,
  aqi: 40, aqiLabel: 'Good',
  hourlyTemp:
[21,20,20,19,19,20,21,22,23,24,25,26,26,25,25,24,23,22,22,21,21,20,20,20],
  hourlyPrecip: [0,0,0,5,5,0,0,0,0,0,5,10,15,10,5,0,0,0,0,5,10,5,0,0],
  hourlyHumidity:
[65,68,70,72,70,66,62,60,57,55,54,56,58,60,62,63,62,60,62,65,67,68,66,65],
  aqiBreakdown: [12,18,15,10,8,6],
  forecast: [
    {day:'Mon',icon:'🌤️',high:25,low:19},{day:'Tue',icon:'☀️',high:27,low:20},
    {day:'Wed',icon:'☀️',high:28,low:21},{day:'Thu',icon:'☁️',high:24,low:18},
    {day:'Fri',icon:'🌧️',high:21,low:16},{day:'Sat',icon:'☁️',high:23,low:17},
    {day:'Sun',icon:'☀️',high:26,low:20}
  ]
 }
};

const hours = Array.from({length:24}, (_,i) => `${String(i).padStart(2,'0')}:00`);
const aqiLabels = ['PM2.5','PM10','O₃','NO₂','SO₂','CO'];

let tempChart, precipChart, aqiRadarChart, aqiDoughnutChart;
let currentCity = 'mumbai';
```

```
const chartDefaults = {
  responsive: true,
  maintainAspectRatio: false,
  plugins: { legend: { labels: { color: '#94a3b8', font: { size: 11 } } } },
  scales: {
    x: { ticks: { color: '#475569', font: { size: 10 } }, grid: { color:
'rgba(255,255,255,0.04)' } },
    y: { ticks: { color: '#475569', font: { size: 10 } }, grid: { color:
'rgba(255,255,255,0.04)' } }
  }
};

function createGradient(ctx, c1, c2) {
  const g = ctx.createLinearGradient(0, 0, 0, 260);
  g.addColorStop(0, c1);
  g.addColorStop(1, c2);
  return g;
}

function initCharts(data) {
  const tempCtx = document.getElementById('tempChart').getContext('2d');
  const precipCtx = document.getElementById('precipChart').getContext('2d');
  const radarCtx = document.getElementById('aqiRadar').getContext('2d');
  const doughnutCtx = document.getElementById('aqiDoughnut').getContext('2d');

  tempChart = new Chart(tempCtx, {
    type: 'line',
    data: {
      labels: hours,
      datasets: [{
        label: 'Temperature °C',
        data: data.hourlyTemp,
        borderColor: '#38bdf8',
        backgroundColor: createGradient(tempCtx, 'rgba(56,189,248,0.3)',
'rgba(56,189,248,0)'),
        fill: true, tension: 0.4, pointRadius: 0, pointHoverRadius: 6,
        pointHoverBackgroundColor: '#38bdf8', borderWidth: 2
      }]
    },
    options: { ...chartDefaults, interaction: { intersect: false, mode: 'index' } }
  });

  precipChart = new Chart(precipCtx, {
    type: 'bar',
    data: {
      labels: hours,
      datasets: [
        {
```

```
      label: 'Precipitation %',
      data: data.hourlyPrecip,
      backgroundColor: 'rgba(167,139,250,0.6)',
      borderRadius: 4, barPercentage: 0.6
    },
    {
      label: 'Humidity %',
      data: data.hourlyHumidity,
      type: 'line', borderColor: '#34d399',
      backgroundColor: 'transparent',
      tension: 0.4, pointRadius: 0, borderWidth: 2, yAxisID: 'y1'
    }
  ]
 },
 options: {
  ...chartDefaults,
  scales: {
   ...chartDefaults.scales,
   y1: {
    position: 'right',
    ticks: { color: '#475569', font: { size: 10 } },
    grid: { display: false },
    min: 0, max: 100
   }
  }
 }
});

aqiRadarChart = new Chart(radarCtx, {
 type: 'radar',
 data: {
  labels: aqiLabels,
  datasets: [{
   label: 'AQI Components',
   data: data.aqiBreakdown,
   borderColor: '#f472b6',
   backgroundColor: 'rgba(244,114,182,0.15)',
   pointBackgroundColor: '#f472b6',
   pointRadius: 4, borderWidth: 2
  }]
 },
 options: {
  responsive: true, maintainAspectRatio: false,
  scales: {
   r: {
    grid: { color: 'rgba(255,255,255,0.06)' },
    angleLines: { color: 'rgba(255,255,255,0.06)' },
    ticks: { display: false },
```

```
        pointLabels: { color: '#94a3b8', font: { size: 11 } }
      }
    },
    plugins: { legend: { display: false } }
  }
});

aqiDoughnutChart = new Chart(doughnutCtx, {
  type: 'doughnut',
  data: {
    labels: ['AQI', 'Remaining'],
    datasets: [{
      data: [data.aqi, 500 - data.aqi],
      backgroundColor: [aqiColor(data.aqi), 'rgba(255,255,255,0.04)'],
      borderWidth: 0, cutout: '78%'
    }]
  },
  options: {
    responsive: true, maintainAspectRatio: false,
    plugins: { legend: { display: false }, tooltip: { enabled: false } }
  }
});
}

function aqiColor(val) {
  if (val <= 50) return '#22c55e';
  if (val <= 100) return '#eab308';
  if (val <= 150) return '#f97316';
  return '#ef4444';
}

function uvDescription(uv) {
  if (uv <= 2) return 'Low';
  if (uv <= 5) return 'Moderate';
  if (uv <= 7) return 'High — Wear Sunscreen';
  if (uv <= 10) return 'Very High — Limit Exposure';
  return 'Extreme — Stay Indoors';
}

function updateDashboard(cityKey) {
  const d = CITIES[cityKey];
  currentCity = cityKey;

  document.getElementById('heroTemp').textContent = d.temp + '°';
  document.getElementById('heroCondition').textContent = d.condition;
  document.getElementById('heroIcon').textContent = d.icon;
  document.getElementById('feelsLike').textContent = d.feelsLike + '°';
  document.getElementById('heroHumidity').textContent = d.humidity + '%';
```

```
  document.getElementById('heroPressure').textContent = d.pressure;
  document.getElementById('windSpeed').textContent = d.wind + ' km/h';
  document.getElementById('windDir').textContent = d.windDir + ' — ' + (d.wind <
12 ? 'Light' : d.wind < 20 ? 'Gentle Breeze' : 'Fresh Wind');
  document.getElementById('uvValue').textContent = d.uv;
  document.getElementById('uvLabel').textContent = uvDescription(d.uv);
  document.getElementById('visValue').textContent = d.vis + ' km';
  document.getElementById('rainChance').textContent = d.rain + '%';
  document.getElementById('rainSub').textContent = d.rain > 50 ? 'Heavy rain
likely' : d.rain > 20 ? 'Light showers expected' : 'Unlikely today';
  document.getElementById('aqiLabel').textContent = d.aqiLabel + ' — ' + d.aqi;

  tempChart.data.datasets[0].data = d.hourlyTemp;
  tempChart.update('none');

  precipChart.data.datasets[0].data = d.hourlyPrecip;
  precipChart.data.datasets[1].data = d.hourlyHumidity;
  precipChart.update('none');

  aqiRadarChart.data.datasets[0].data = d.aqiBreakdown;
  aqiRadarChart.update('none');

  aqiDoughnutChart.data.datasets[0].data = [d.aqi, 500 - d.aqi];
  aqiDoughnutChart.data.datasets[0].backgroundColor = [aqiColor(d.aqi),
'rgba(255,255,255,0.04)'];
  aqiDoughnutChart.update('none');

  renderForecast(d.forecast);
}

function renderForecast(days) {
  const row = document.getElementById('forecastRow');
  row.innerHTML = days.map(d => `
    <div class="forecast-day" onmouseenter="showTip(event, '${d.day}:
${d.high}°/${d.low}° ${d.icon}')" onmouseleave="hideTip()">
      <div class="day-name">${d.day}</div>
      <div class="day-icon">${d.icon}</div>
      <div class="day-high">${d.high}°</div>
      <div class="day-low">${d.low}°</div>
    </div>
  `).join('');
}

function showTip(e, text) {
  const tip = document.getElementById('tooltip');
  tip.textContent = text;
  tip.style.opacity = '1';
  tip.style.left = e.pageX - 40 + 'px';
```

```
  tip.style.top = e.pageY - 48 + 'px';
}
function hideTip() {
  document.getElementById('tooltip').style.opacity = '0';
}

function openPopup(type) {
  const d = CITIES[currentCity];
  const overlay = document.getElementById('popupOverlay');
  const title = document.getElementById('popupTitle');
  const body = document.getElementById('popupBody');

  const popups = {
    temperature: {
      title: '🌡 Temperature Details — ' + d.name,
      html: `
        <div class="popup-stats">
          <div class="popup-stat"><div class="popup-stat-label">Current</div><div
class="popup-stat-value">${d.temp}°C</div></div>
          <div class="popup-stat"><div class="popup-stat-label">Feels Like</div><div
class="popup-stat-value">${d.feelsLike}°C</div></div>
          <div class="popup-stat"><div class="popup-stat-label">Day High</div><div
class="popup-stat-value">${Math.max(...d.hourlyTemp)}°C</div></div>
          <div class="popup-stat"><div class="popup-stat-label">Day Low</div><div
class="popup-stat-value">${Math.min(...d.hourlyTemp)}°C</div></div>
        </div>
        <canvas id="popupChart" height="180"></canvas>
      `,
      chart: ctx => new Chart(ctx, {
        type: 'line',
        data: {
          labels: hours.filter((_,i) => i%3===0),
          datasets: [{
            data: d.hourlyTemp.filter((_,i) => i%3===0),
            borderColor: '#38bdf8', backgroundColor: 'rgba(56,189,248,0.1)',
            fill: true, tension: 0.4, pointRadius: 4, pointBackgroundColor: '#38bdf8'
          }]
        },
        options: { ...chartDefaults, plugins: { legend: { display: false } } }
      })
    },
    wind: {
      title: '🌬 Wind Analysis — ' + d.name,
      html: `
        <div class="popup-stats">
          <div class="popup-stat"><div class="popup-stat-label">Speed</div><div
class="popup-stat-value">${d.wind} km/h</div></div>
```

```
        <div class="popup-stat"><div class="popup-stat-label">Direction</div><div
class="popup-stat-value">${d.windDir}</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Gusts Up
To</div><div class="popup-stat-value">${d.wind + 8} km/h</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Beaufort
Scale</div><div class="popup-stat-value">${d.wind < 12 ? '2 — Light' : d.wind < 20
? '3 — Gentle' : '4 — Moderate'}</div></div>
      </div>
      <p style="color:var(--text-dim);font-size:13px;margin-top:8px">
        ${d.wind > 20 ? '⚠ Moderate winds may affect outdoor activities. Secure
loose objects.' : 'Conditions are calm and comfortable for outdoor activities.'}
      </p>
    `
  },
  uv: {
    title: '☀ UV Index — ' + d.name,
    html: `
      <div class="popup-stats">
        <div class="popup-stat"><div class="popup-stat-label">Current
UV</div><div class="popup-stat-value" style="color:${d.uv > 7 ? '#ef4444' : d.uv >
5 ? '#f97316' : '#eab308'}">${d.uv}</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Category</div><div
class="popup-stat-value" style="font-size:16px">${uvDescription(d.uv).split('—
')[0].trim()}</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Peak Hour</div><div
class="popup-stat-value">12:00–14:00</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Protection</div><div
class="popup-stat-value" style="font-size:14px">${d.uv > 5 ? 'SPF 30+' : 'SPF
15'}</div></div>
      </div>
      <div style="margin-
top:12px;padding:14px;background:rgba(244,114,182,0.08);border-
radius:10px;border:1px solid rgba(244,114,182,0.2);font-size:13px;color:var(--text-
dim)">
        ${d.uv >= 6 ? '🧴 <strong style="color:var(--
text)">Recommendation:</strong> Apply broad-spectrum sunscreen. Wear
sunglasses and a hat between 11am–3pm.' :
          '✅ <strong style="color:var(--text)">Low Risk:</strong> Normal outdoor
activity is safe. Sunscreen still recommended for extended exposure.'}
      </div>
    `
  },
  visibility: {
    title: '◎ Visibility — ' + d.name,
    html: `
      <div class="popup-stats">
```

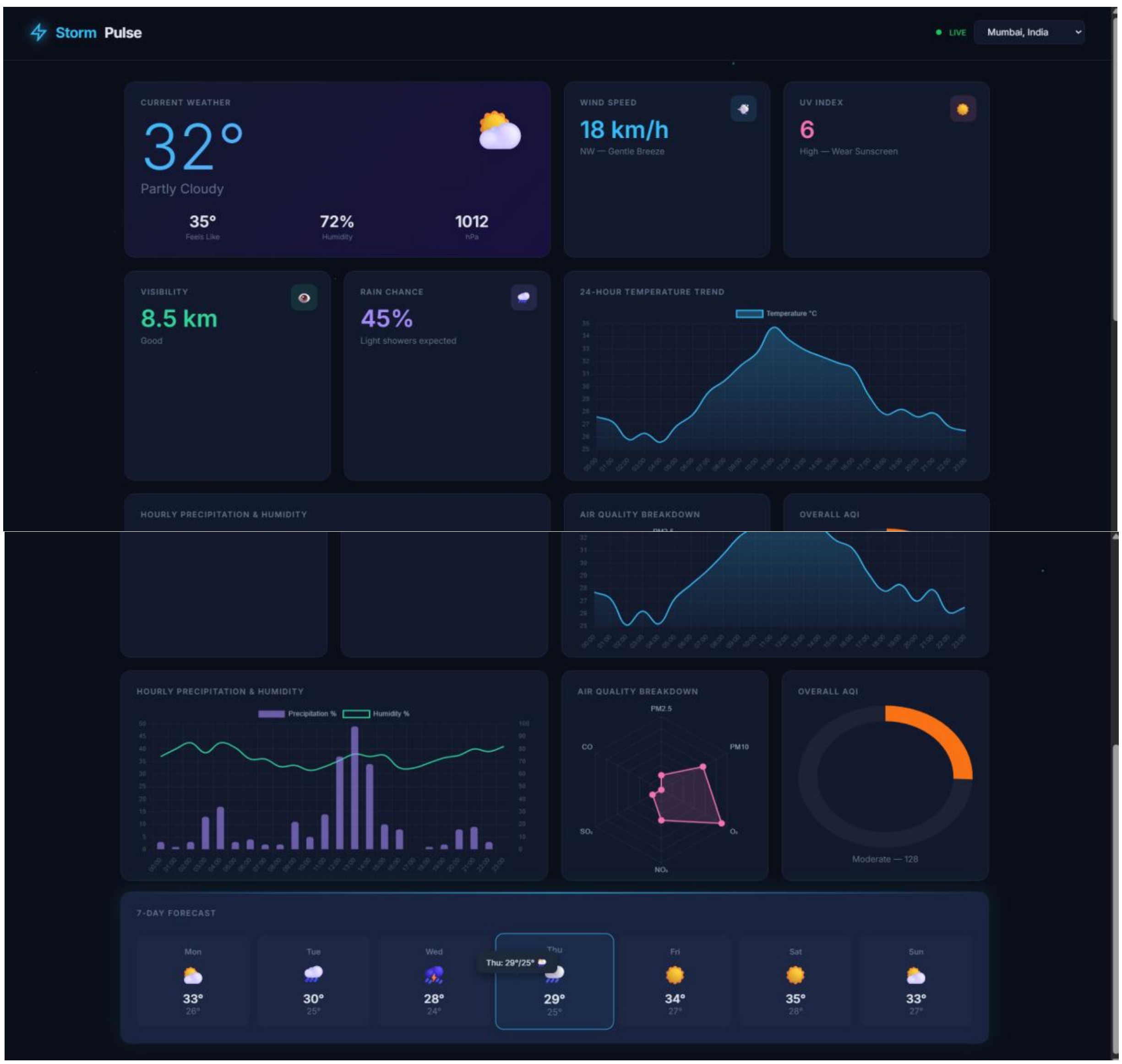
```
        <div class="popup-stat"><div class="popup-stat-label">Distance</div><div class="popup-stat-value">${d.vis} km</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Rating</div><div class="popup-stat-value">${d.vis > 10 ? 'Excellent' : d.vis > 5 ? 'Good' : 'Poor'}</div></div>
      </div>
      <p style="color:var(--text-dim);font-size:13px;margin-top:8px">
        ${d.vis < 5 ? '⚠️ Reduced visibility. Drive with caution and use headlights.' : 'Clear conditions with good horizontal visibility.'}
      </p>
    `
  },
  rain: {
    title: '🌧️ Precipitation — ' + d.name,
    html: `
      <div class="popup-stats">
        <div class="popup-stat"><div class="popup-stat-label">Chance</div><div class="popup-stat-value">${d.rain}%</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Type</div><div class="popup-stat-value">${d.rain > 50 ? 'Rain' : 'Drizzle'}</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Peak Hour</div><div class="popup-stat-value">${hours[d.hourlyPrecip.indexOf(Math.max(...d.hourlyPrecip))]}</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Accumulation</div><div class="popup-stat-value">${(d.hourlyPrecip.reduce((a,b)=>a+b,0)/100*2.5).toFixed(1)} mm</div></div>
      </div>
      <canvas id="popupChart" height="180"></canvas>
    `,
    chart: ctx => new Chart(ctx, {
      type: 'bar',
      data: {
        labels: hours.filter((_,i) => i%2===0),
        datasets: [{
          data: d.hourlyPrecip.filter((_,i) => i%2===0),
          backgroundColor: 'rgba(167,139,250,0.5)', borderRadius: 4
        }]
      },
      options: { ...chartDefaults, plugins: { legend: { display: false } } }
    })
  },
  aqi: {
    title: '🫁 Air Quality — ' + d.name,
    html: `
      <div class="popup-stats">
```

```
        <div class="popup-stat"><div class="popup-stat-label">AQI</div><div
class="popup-stat-value" style="color:${aqiColor(d.aqi)}">${d.aqi}</div></div>
        <div class="popup-stat"><div class="popup-stat-label">Category</div><div
class="popup-stat-value" style="font-size:16px">${d.aqiLabel}</div></div>
        ${aqiLabels.map((l,i) => `<div class="popup-stat"><div class="popup-stat-
label">${l}</div><div class="popup-stat-
value">${d.aqiBreakdown[i]}</div></div>`).join('')}
      </div>
      <div style="margin-
top:12px;padding:14px;background:rgba(52,211,153,0.08);border-
radius:10px;border:1px solid rgba(52,211,153,0.2);font-size:13px;color:var(--text-
dim)">
        ${d.aqi > 100 ? '😷 <strong style="color:var(--text)">Advisory:</strong>
Sensitive groups should reduce prolonged outdoor exertion.' :
         '✅ <strong style="color:var(--text)">All Clear:</strong> Air quality is
acceptable for all activities.'}
      </div>
    `
   }
  };

  const p = popups[type];
  title.textContent = p.title;
  body.innerHTML = p.html;
  overlay.classList.add('active');

  if (p.chart) {
    requestAnimationFrame(() => {
      const canvas = document.getElementById('popupChart');
      if (canvas) p.chart(canvas.getContext('2d'));
    });
  }
}

function closePopup(e) {
  if (e && e.target !== e.currentTarget) return;
  document.getElementById('popupOverlay').classList.remove('active');
}

document.addEventListener('keydown', e => { if (e.key === 'Escape') closePopup();
});

function showToast(msg) {
  const toast = document.getElementById('toast');
  document.getElementById('toastText').innerHTML = msg;
  toast.classList.add('show');
  setTimeout(() => toast.classList.remove('show'), 4500);
}
```

```
function createParticles() {
  const container = document.getElementById('particles');
  for (let i = 0; i < 30; i++) {
    const p = document.createElement('div');
    p.className = 'particle';
    const size = Math.random() * 3 + 1;
    p.style.width = size + 'px';
    p.style.height = size + 'px';
    p.style.left = Math.random() * 100 + '%';
    p.style.animationDuration = (Math.random() * 15 + 10) + 's';
    p.style.animationDelay = (Math.random() * 10) + 's';
    container.appendChild(p);
  }
}

document.getElementById('citySelect').addEventListener('change', function() {
  updateDashboard(this.value);
  showToast(`<strong>📍 Switched to ${CITIES[this.value].name}</strong>Data refreshed with latest readings`);
});

createParticles();
initCharts(CITIES.mumbai);
renderForecast(CITIES.mumbai.forecast);

const alerts = [
  '<strong>⚡ Weather Alert</strong>UV index rising — stay hydrated!',
  '<strong>🌡️ Heat Advisory</strong>Temperature spike expected at noon',
  '<strong>🌧️ Rain Incoming</strong>Carry an umbrella this evening',
  '<strong>🌬️ Wind Advisory</strong>Gusts up to 35 km/h after 4pm'
];
let alertIdx = 0;
setTimeout(() => {
  showToast(alerts[0]);
  setInterval(() => {
    alertIdx = (alertIdx + 1) % alerts.length;
    showToast(alerts[alertIdx]);
  }, 20000);
}, 3000);

setInterval(() => {
  const d = CITIES[currentCity];
  const jitter = () => (Math.random() - 0.5) * 2;
  d.hourlyTemp = d.hourlyTemp.map(t => +(t + jitter() * 0.3).toFixed(1));
  d.hourlyPrecip = d.hourlyPrecip.map(p => Math.max(0, Math.min(100, +(p + jitter() * 2).toFixed(0))));
```

```
  d.hourlyHumidity = d.hourlyHumidity.map(h => Math.max(30, Math.min(100, +(h
+ jitter()).toFixed(0))));

  tempChart.data.datasets[0].data = d.hourlyTemp;
  tempChart.update('none');
  precipChart.data.datasets[0].data = d.hourlyPrecip;
  precipChart.data.datasets[1].data = d.hourlyHumidity;
  precipChart.update('none');
}, 5000);
</script>
</body>
</html>
```

Output:



Conclusion:
In this assignment, an interactive data visualization dashboard was successfully designed and developed using HTML, CSS, and JavaScript. The use of charting libraries like Chart.js or D3.js enabled effective representation of data through visually appealing charts and graphs.
The integration of Bootstrap ensured a responsive and structured layout, enhancing user experience across devices. This project provides practical knowledge of how data can be analyzed and presented in a meaningful way, which is essential in real-world applications. Overall, the assignment strengthens skills in front-end development, data visualization, and interactive design.